

UTSAV DARLAMI

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EXPERIENCE

Graduate Research Assistant

Department of Computer Science, University of Alabama in Huntsville

August 2025 - Present

Huntsville, Alabama

- Conducting research on interaction patterns and view management in multi-display environments.
- Developed **CanvasChat**, an intelligent system using **React Flow** and **LLMs** to assist with spatial information organization.

Machine Learning Engineer (Full Stack)

Docsumo

September 2021 - April 2025

Kathmandu, Nepal

- **Lead Developer** for a Python package streamlining LLM-based information extraction using **LiteLLM**, **Langfuse**, and **Pydantic**; this library optimized experimentation and powered the company's fully automated information extraction API.
- Designed and built an LLM-based table extraction pipeline to convert unstructured text into structured data; benchmarked performance against existing solutions using models like GPT-4o, Mixtral-8x7B, and Llama 3.1, achieving **75%** accuracy with **GPT-4o** in production and scaling to process an average of **300+** documents for 2 enterprise clients.
- **Slashed processing time by 60%** by engineering a data featurization pipeline for specialized **scikit-learn** models (identifying column labels, valid rows) from OCR data.
- Collaborated directly with clients to define requirements and deploy custom extraction logic specific to their business use cases.
- Built and deployed a scalable text classification API using **Fasttext**, **SpaCy**, **FastAPI**, and **GCP**, which currently supports four downstream services.

PROJECTS

CanvasChat

Python, Gemini, Google ADK, React Flow, WebSockets

[Github](#)

- Built LLM driven workspace that organizes charts, documents, and images across displays from freeform commands, shifting interaction from manual arranging to semantic grouping and cross display layouts.
- Implemented backend semantic layer and 4 stage agent pipeline (extraction, context assembly, intent to plan, layout execution) using Gemini and Google ADK with reviewable plans and history revert.
- Developed shared React Flow canvas with real time WebSocket sync, chat panel, and visual overlays, preserving direct manipulation for fine tuning.

Agent Starter Template (Google ADK)

Python, FastAPI, Google ADK, JavaScript

[Github](#)

- Built a stateful AI agent architecture using FastAPI and the Google ADK, including a JavaScript client for real-time state updates.
- Created a tool-calling system with Pydantic to let the AI safely execute functions.

Evolutionpy

Python, NumPy, Meta-heuristic Algorithms, OOP

[Github](#)

- Built a Python library for evolutionary optimization to solve optimization problems.
- Designed a modular codebase using Object-Oriented Programming (OOP) to easily support custom fitness functions.

Sandstone Microscopic Image Segmentation

Python, Scikit-learn, Random Forest, Image Processing

[Github](#)

- Generated feature sets from microscopic images using filters like Gaussian, Sobel, and Gabor.
- Trained a Random Forest classifier on these features to identify mineral segments (Quartz, Pore, Clay), achieving a **95%** F1-Score (Dice Coefficient) on the segmentation task.

BreakfastScoop

Python, BeautifulSoup, Flask, NLP, Machine Learning

[Github](#)

- Developed a Flask application that fetches live news by scraping multiple Nepali websites with a custom BeautifulSoup script, and categorizes it using the trained model.
- Categorized articles into 10 topics using a Naive Bayes classifier with an **81%** F1-score.

SKILLS

Programming Languages: Python, C/C++, Bash, JavaScript

Databases & Tools: SQL, MongoDB, Docker, Git, Postman, Linux, GCP (Cloud Run, Buckets)

Machine Learning & Data Science: PyTorch, Scikit-learn, OpenCV, spaCy, fasttext, Transformers

LLM Frameworks: LangChain, LiteLLM, Google ADK

Languages: English, Nepali, Hindi

PUBLICATIONS AND PRESENTATIONS

Publication: [Microplastic Contamination in High-Altitude Soils of Sagarmatha National Park](#) in *Microplastics* 2026
Baniya, S., Mohanta, T., Yacoub, M., **Darlamī, U.**, Li, A., Subedi, I., Nicholson, K., Sharma, S., and Han, B.

Presentation: “ANN-GA for Optimization of Multivariate Function” at NCMA 2022, Nepal June 2022

EDUCATION

Master of Science in Computer Science

University of Alabama in Huntsville, Huntsville, Alabama

August 2025 - May 2027 (Expected)

Bachelor of Science in Computer Science

Kathmandu University, Dhulikhel, Nepal

August 2017 - March 2022